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1. The *Catalan numbers* are given by

$$C_n = \frac{1}{n+1} \binom{2n}{n}$$

for any integer $n \geq 0$. (Recall that $\binom{0}{0} = 1$ – there’s only one way to choose no elements from an empty set.) Prove that

$$C_n = \frac{2(2n-1)}{n+1} C_{n-1}$$

for all $n \geq 1$, and hence that $C_n \leq 4^n$ for all $n \geq 0$.

2. Let m, n be positive integers. A function $f : \{1, \dots, m\} \rightarrow \{1, \dots, n\}$ is said to be *weakly increasing* if $x < y$ implies $f(x) \leq f(y)$. For fixed m, n , how many weakly increasing functions are there? Write your answer as an expression involving m, n and justify how you derived that expression.